

Thm. Galois Theory

Let K be a finite normal extension of a field F , with $G(K/F)$.

There exists a bijective map

$$\lambda: \{E \mid F \leq E \leq K\} \rightarrow \{G(K/E) \mid G(K/E) \leq G(K/F)\}$$

$$: E \mapsto G(K/E)$$

satisfying the following conditions:

$$1. E = K_{G(K/E)} = K^{\lambda(E)}$$

$$2. \text{ For } H \leq G(K/F), \lambda(E_H) = H$$

$$3. [K:E] = |\lambda(E)| \text{ and } [E:F] = (G(K/F) : G(K/E)).$$

$$4. E \text{ is Normal Extension of } F \text{ if and only if } G(K/E) \trianglelefteq G(K/F).$$

$$\text{In this case, } G(E/F) \cong G(K/F) / G(K/E)$$

Proof of 1. $E \leq K_{G(K/E)}$ is trivial. Conversely, suppose $K_{G(K/E)} \not\subseteq E$.

Let $\alpha \in K_{G(K/E)}$ but $\alpha \notin E$. Since $\alpha \notin E$, $\text{irr}(\alpha, F)$ have distinct roots of more than 2, because K is separable. Now, we can construct a conjugation $\tau_{\alpha, \alpha'}$ leaving E fixed and mapping α to α' , the different root of $\text{irr}(\alpha, F)$.

It is contradiction

Proof of 2. Let $H \leq G(K/F)$ be a subgroup. Clearly,

$$H \leq \lambda(K_H) = G(K/K_H) \leq G(K/F)$$

suppose that $H < \lambda(K_H)$. Since K is finite separable extension of F , so is of K_H , thus $K = K_H(\alpha)$ for some $\alpha \in K$. Let

$$n = [K:K_H] = [K:K_H] = |G(K/K_H)|.$$

Then, $H < G(K/K_H)$ implies that $|H| < n$. Denote $H = \{\sigma_i \mid 1 \leq i \leq |H|\}$.

Define

$$f(x) = \prod_{i=1}^{|H|} (x - \sigma_i(\alpha)).$$

Then, each coefficient of f are contained in K_H , because these are symmetric,

And, $f(\alpha) = 0$, because 1 , identity is contained in H . Now,

$$\deg(\alpha, K_H) \leq |H| < n = [K_H(\alpha):K_H]$$

This is contradiction

Lemma. Let K/F be a finite normal extension, and let E be an intermediate, $F \subseteq E \subseteq K$.
 Then, K is normal of E , and $G(K/E) \leq G(K/F)$. Moreover, for any $\sigma, \tau \in G(K/F)$,
 $\sigma|_E = \tau|_E$ if and only if $\sigma \cdot G(K/E) = \tau \cdot G(K/E)$.

Proof. Since K/F is finite, K/E is separable. And, if K is splitting field of $\{f_i(x)/i\}$, it is also polynomials of E , thus proved.

Thm. Let K be a finite normal extension of F , and $F \subseteq E \subseteq K$.
 Then K is a finite normal extension of E and

$$\text{Aut}(K/E) \leq \text{Aut}(K/F).$$

K
 $\uparrow$
 E
 $\uparrow$
 F

↖ some.

Moreover, two Automorphisms σ and τ in $\text{Aut}(K/F)$ are satisfied $\sigma|_E = \tau|_E$
 if and only if

$$\sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E)$$

Proof. Above two assertions are trivial. Let $\sigma, \tau \in \text{Aut}(K/F)$ be given.

$$\begin{aligned} \sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E) &\Leftrightarrow \tau^{-1}\sigma = \mu \text{ for some } \mu \in \text{Aut}(K/E) \\ &\Leftrightarrow \sigma|_E = \tau|_E \end{aligned}$$

Proof of 4. Since E is separable over F , being intermediate field of finite extension, thus E/F is Normal iff E/F is splitting field.

And, E/F is splitting iff every automorphism of K leaving F fixed maps E onto itself
~~because K/F is splitting~~. That is, E/F is splitting if and only if

$$\text{for any } \sigma \in G(K/F) \text{ and } \alpha \in E, \sigma(\alpha) \in E$$

and since $E = K_{G(K/E)}$, by the first assertion,

$$\sigma(\alpha) \in E \Leftrightarrow \text{for all } \tau \in G(K/E), (\tau \circ \sigma)(\alpha) = \sigma(\alpha).$$

That is, $(\sigma^{-1} \circ \tau \circ \sigma)(\alpha) = \alpha$. This means $\sigma^{-1} \circ \tau \circ \sigma \in G(K/E)$.

Finally, $\varphi: G(K/F) \rightarrow G(E/F): \sigma \mapsto \sigma|_E$ is onto, by E is Normal, and

$$\ker \varphi = G(K/E). \text{ Thus, } G(E/F) \cong G(K/F) / G(K/E).$$

Recall Note.

E/F is finite extension $\Leftrightarrow E = F(\alpha_1, \dots, \alpha_n)$ for some algebraic $\alpha_i \in \overline{F}$.

In this case,

$$\begin{aligned} [E:F] &= [E:F(\alpha_1, \dots, \alpha_{n-1})][F(\alpha_1, \dots, \alpha_{n-1}):F(\alpha_1, \dots, \alpha_{n-2})] \cdots [F(\alpha_1):F] \\ &= \deg \text{irr}(\alpha_n, F(\alpha_1, \dots, \alpha_{n-1})) \cdots \deg \text{irr}(\alpha_1, F) \\ &= \prod_{i=1}^n \deg \text{irr}(\alpha_i, F). \end{aligned}$$

$$\begin{aligned} \{E:F\} &\stackrel{\text{def}}{=} \# \{ \varphi: E \rightarrow \overline{F} \mid \varphi \text{ is Homomorphism, leaving } F \text{ fixed} \} \\ &= \{E:F(\alpha_1, \dots, \alpha_{n-1})\} \cdots \{F(\alpha_1):F\} \\ &= \prod_{i=1}^n \# \{ x \in \overline{F} \mid \text{irr}(\alpha_i, F(\alpha_0, \alpha_1, \dots, \alpha_{i-1}))(\varphi) = 0 \} \quad (\alpha_0 = 1) \\ &= \prod_{i=1}^n \text{number of conjugate of } \alpha_i \text{ over } F(\alpha_0, \dots, \alpha_{i-1}) \end{aligned}$$

Lemma. If K is finite Normal extension of F , then

$F \leq E \leq K \Rightarrow K$ is Normal extension of E .

Def. A polynomial $f(x) \in F[x]$ is called separable if each irreducible factors have no multiple root.

A field in which every polynomial is separable are called perfect.

An extension E/F , and $\alpha \in E$ is called separable if α is transcendental or $\text{irr}(\alpha, F)$ is separable

and E/F is called separable if all elements are separable.

+ $\{F(\alpha):F\} = \# \text{distinct zeros of } \text{irr}(\alpha, F)$, because

if $\varphi: F(\alpha) \rightarrow \overline{F}$ leaves F fixed, then

$$0 = \varphi(0) = \varphi(\text{irr}(\alpha, F)(\alpha)) = \text{irr}(\alpha, F)(\varphi(\alpha))$$

Therefore, $\varphi(\alpha)$ must be a root of $\text{irr}(\alpha, F)$.

And, for each root β of $\text{irr}(\alpha, F)$, conjugate iso. $\varphi_{\alpha, \beta}$ exists.

Irreducible.

Lemma. Let $\varphi: R \rightarrow S$ be a Ring Homomorphism of Commutative Ring R, S with 1.

Then, $\varphi^*: R[x] \rightarrow S[x]; \sum r_i x^i \mapsto \sum \varphi(r_i) \cdot x^i$ is also Ring Homomorphism.

Proof. Fucking trivial.

Thm. Let R be an integral domain, and F be a field, $\varphi: R \rightarrow F$ be a Ring Homomorphism.

Given $p(x) \in R[x]$, if $\deg \varphi^*(p) = \deg p$ and $\varphi^*(p)$ is irreducible in $F[x]$, then there are no polynomial $f, g \in R[x]$ s.t. $p = f \cdot g$ and $\deg g, \deg f < \deg p$.

Proof. Under the condition above, suppose $p(x) = f(x) \cdot g(x)$ with $\deg f < \deg p, f, g \in R[x]$

Then, $\varphi^*(p) = \varphi^*(f) \cdot \varphi^*(g)$. Since $\varphi^*(p)$ is irreducible, either $\varphi^*(f)$ or $\varphi^*(g)$ degree 0.

If $\varphi^*(f) = 0$, then

$$\deg p = \deg \varphi^*(p) = \deg \varphi^*(f) + \deg \varphi^*(g) = \deg \varphi^*(g) \leq \deg g < \deg p$$

which is contradiction, and similarly in case of $\varphi^*(g) = 0$.

Def. $f(x) = a_0 + a_1 x + \dots + a_n x^n \in \mathbb{Z}[x]$ is called primitive if $(a_0, a_1, \dots, a_n) = (1)$.

Lemma. Product of two primitive polynomials is primitive.

Proof. If $f(x)g(x)$ is not primitive where $f(x)$ and $g(x)$ are primitive, then for some prime p , p divides all coefficients of $f(x)g(x)$. Let $\varphi: \mathbb{Z} \rightarrow \mathbb{Z}_p; a \mapsto a \bmod p$. Then,

$$\varphi^*(fg) = \varphi^*(f) \cdot \varphi^*(g) = 0 \text{ in } \mathbb{Z}_p[x].$$

But, both $\varphi^*(f)$ and $\varphi^*(g)$ are not zero, it is contradiction to $\mathbb{Z}_p[x]$ is domain.

Lemma. Every nonzero $f(x) \in \mathbb{Q}[x]$ has a unique factorization

$$f(x) = c(f) \cdot f^*(x)$$

where $c(f) \in \mathbb{Q}$ is positive, $f^*(x) \in \mathbb{Z}[x]$ is primitive.

Proof. Let $f(x) = \frac{a_0}{b_0} + \frac{a_1}{b_1}x + \dots + \frac{a_n}{b_n}x^n \in \mathbb{Q}[x]$. Put $B = b_0 \dots b_n$. So, $Bf(x) \in \mathbb{Z}[x]$.

Take $D = \pm d$, where d is gcd of $(a_0, \dots, a_n)$, and the sign is determined to make $\frac{D}{B}$ be positive.

Now, $\frac{D}{B} \cdot f(x) = D^{-1} \cdot Bf(x) = D^{-1}g(x) \in \mathbb{Z}[x]$ and it is primitive. Now, put

$c = D/B$ and $f^* = (B/D)f$, then we obtain that we desired. To show that uniqueness,

suppose $f(x) = c(f) \cdot f^*(x) = c' h(x)$, $c \in \mathbb{Q}^+$, $h \in \mathbb{Z}[x]$ is primitive. Now, $f^*(x) = (c/c(f)) \cdot h(x)$.

Write $c/c(f) = u/v$, where $(u, v) = 1$. Then, $v f^*(x) = u h(x)$, now $v = u = 1$, because

v is common divisor of coefficients of $u h(x)$, but $h(x)$ is primitive.

Corollary. If $f(x) \in \mathbb{Q}[x]$ factors as $f(x) = g(x)h(x)$ in $\mathbb{Q}[x]$, then

$$c(f) = c(g) \cdot c(h) \quad \text{and} \quad f^{\#}(x) = g^{\#}(x) \cdot h^{\#}(x).$$

Proof. $f_G = g_G h_G = c(g) \cdot g^{\#}_G \cdot c(h) \cdot h^{\#}_G$.

Gauss Thm. If $p(x) \in \mathbb{Z}[x]$ is not a product of two polynomials in $\mathbb{Z}[x]$ each $< \deg p$, then $p(x)$ is irreducible in $\mathbb{Q}[x]$.

Lemma. Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F .

Then, for any $\sigma \in \text{Aut}(K/F)$, $\sigma(\alpha)$ is a root of minimal polynomial for α over F .
(Equivalently, any polynomial with coefficients in F having α as a root also has $\sigma\alpha$ as a root.)

Proof. Suppose $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ ($a_i \in F$) and

$$f(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0 = 0.$$

$$\begin{aligned}\text{Then, } 0 &= \sigma(f(\alpha)) = \sigma(\alpha^n) + \sigma(a_{n-1}\alpha^{n-1}) + \dots + \sigma(a_1\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1\sigma\alpha + a_0 \\ &= f(\sigma\alpha)\end{aligned}$$

Thus proved.

Cyclotomic Extensions.

Def. The splitting field of $x^n - 1$ over F is called the n th cyclotomic extension of F .

And, $\Phi_n(x) \stackrel{\text{def}}{=} \prod_{i=1}^{\phi(n)} (x - \alpha_i)$ where α_i are the primitive n th roots of unity in $\overline{F}$,
is called the n th cyclotomic polynomial over F .

Lemma. If p is prime number, then

$$\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + x + 1 = \frac{x^p - 1}{x - 1}$$

is irreducible over $\mathbb{Q}$.

Proof. Note: $x^n - y^n = (x - y) \cdot (x^{n-1} + x^{n-2}y + \dots + xy^{n-2} + y^{n-1}) = (x - y) \cdot \left(\sum_{k=0}^{n-1} x^{n-1-k} y^k \right)$
Since $\Phi_p(x)$ is irreducible $\Leftrightarrow \Phi_p(x+1)$ is irreducible, because

$$\Phi_p(x) = f(x) \cdot g(x) \Leftrightarrow \Phi_p(x+1) = f(x+1) \cdot g(x+1).$$

And, since $\Phi_p(x+1) = (x+1)^p - 1 = x^p + \binom{p}{1}x^{p-1} + \dots + \binom{p}{p-1}x + 1 - 1,$

it is irreducible by the Eisenstein's criterion.

$$\begin{aligned} \Phi_8(x) &= (x - \zeta)(x - \zeta^3)(x - \zeta^5)(x - \zeta^7) \\ &= (x - \zeta)(x + \zeta)(x - \zeta^3)(x + \zeta^3) \\ &= (x^2 - \zeta^2)(x^2 - \zeta^6) \\ &= x^4 - (\zeta^2 + \zeta^6)x^2 + \zeta^8 \\ &= x^4 - (\zeta^2 - \zeta^2)x^2 + 1 \\ &= x^4 + 1 \end{aligned}$$

Galois group of $x^{20} - 1 \in \mathbb{Q}[x]$.

$$\varphi(20) = 8, \quad \text{Gal}(\mathbb{Q}(\zeta_{20})/\mathbb{Q}) \cong \mathbb{Z}_{20}^\times = \mathbb{Z}_8.$$

Thm. Let F be a field of characteristic not dividing n . Then,

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Proof. If $\zeta \in F$ is a n th primitive root, then for $d|n$, $\zeta^{\frac{n}{d}}$ is d th primitive root, because $(\zeta^{\frac{n}{d}})^d = 1$, and $\{\zeta^{\frac{n}{d}}, \zeta^{\frac{2n}{d}}, \dots\}$ are distinct. Conversely, if ζ^i generates subgroup of order d , then it is d th primitive root.

That is, all primitive d th roots for $d|n$ are primitive n th roots, thus we obtain the -.

Over $\mathbb{Q}$,

$$\Phi_1 = x - 1 \quad (x^3 - 1)(x^3 + 1)$$

$$\Phi_2 = x + 1$$

$$\Phi_3 = x^2 + x + 1$$

$$\Phi_4 = (x^4 - 1) / ((x - 1)(x + 1)) = x^2 + 1$$

$$\Phi_5 = x^4 + x^3 + \dots + 1$$

$$\Phi_6 = (x^6 - 1) / ((x - 1)(x + 1)(x^2 + x + 1)) = x^2 - x + 1.$$

$$\Phi_{2n}(x) = \prod_{i=1}^{n-1} (x - \alpha_i) \quad \text{where } \alpha_i \text{ is a primitive root of } x^{2n} - 1.$$

$$\text{Since } x^{2n} - 1 = (x^n - 1)(x^n + 1), \text{ if}$$

Let ζ be a primitive n th root of unity. It generates a set of roots of $x^n - 1$.

Since n is odd, $(-\zeta)^{2n} = 1$, and $2n$ is the least integer s.t. $(-\zeta)^{2r} = 1$.

because $\zeta, \dots, \zeta^{n-1}$ are not 1, $-\zeta, \dots, -\zeta^{n-1}$ so is.

Therefore, if ζ is primitive root of Φ_n , then $-\zeta$ is primitive root of Φ_{2n} .

Thus $\Phi_n(-x)$ divides $\Phi_{2n}(x)$. Meanwhile, $\varphi(n) = \varphi(2n)$, thus these are same.

($\gcd(k, n) = 1 \Leftrightarrow \gcd(k, 2n) = 1$, since n is odd).

The primitive n th root of unity is $\exp(2\pi i \cdot \frac{k}{n})$, where $0 \leq k \leq n-1$, $\gcd(k, n) = 1$.

Meanwhile, for an integer d s.t. $d|n$, d th primitive root is $\exp(2\pi i \cdot \frac{f}{d})$, where $0 \leq f \leq d-1$, $\gcd(f, d) = 1$.

Since $d|n$, if $\gcd(k, n) = 1$, then $\gcd(k, d) = 1$, therefore

$$\exp(2\pi i \cdot \frac{k}{d}) = (\exp(2\pi i \cdot \frac{k}{n}))^{\frac{n}{d}} \text{ is also primitive } d\text{th root. } (\dagger \gcd(n, \frac{n}{d})$$

Lemma. Let $\mathbb{Q}: F \rightarrow F'$ be a field Isomorphism, and let $\mathbb{Q}^*: F[x] \rightarrow F'[x]: \sum r_i x^i \mapsto \sum \mathbb{Q}(r_i) x^i$.
 Let $p(x) \in F[x]$ be irreducible, and let $p^*(x) = \mathbb{Q}^*(p(x))$.
 If β is a root of $p(x)$ and β' is a root of $p^*(x)$,
 then, there exists a unique Isomorphism $\overline{\mathbb{Q}}: F(\beta) \rightarrow F(\beta')$ s.t. $\overline{\mathbb{Q}}|_F = \mathbb{Q}$, $\overline{\mathbb{Q}}(\beta) = \beta'$.
 Proof. Since $\mathbb{Q}^*$ is also Isomorphism,

$$\mathbb{I}: F[x]/(p(x)) \rightarrow F[x]/(p^*(x)): f(x) + (p(x)) \mapsto \mathbb{Q}^*(f(x)) + (p^*(x))$$

It maps $c + (p(x)) \mapsto \mathbb{Q}(c) + (p^*(x))$ for all $c \in F$ and $x + (p(x)) \mapsto x + (p^*(x))$, and Isomorphism.

$$\text{Now, } \overline{\mathbb{Q}}: F(\beta) \xrightarrow{\phi^{-1}} F[x]/(p(x)) \xrightarrow{\mathbb{I}} F[x]/(p^*(x)) \xrightarrow{\phi'} F(\beta')$$

$$: \alpha \mapsto (\phi' \circ \mathbb{I} \circ \phi^{-1})(\alpha)$$

'is a field isomorphism leaving F fixed, and β maps to β' .

Lemma. Let $f(x) \in F[x]$ and $D_x f(x) \in F[x]$ be a derivative of $f(x)$. Then,

$f(x)$ has a multiple root α 'if and only if' $D_x f(x)$ has α as a root.

In particular,

$f(x)$ is separable 'if and only if' $(f(x), D_x f(x)) = 1$.

Proof. Suppose that $(x-\alpha)^2$ divides $f(x)$ in the splitting field K of f . Then,

$$f(x) = (x-\alpha)^2 \cdot h(x) \quad \text{for some } h(x) \in K[x].$$

By the property of derivative,

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot h(x) + (x-\alpha)^2 \cdot D_x h(x) = (x-\alpha)(2h(x) + (x-\alpha) \cdot D_x h(x)).$$

Thus proved. Conversely, suppose that both f and $D_x f$ has α as a root. Then,

$$f(x) = (x-\alpha) \cdot h(x) \quad \text{for some } h(x) \in K[x], \text{ and } D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x).$$

And, by the assumption, $(x-\alpha)$ divides $D_x f$, so divides $h(x)$. Now,

$$f(x) = (x-\alpha)h(x) = (x-\alpha)^2 r(x), \quad \text{where } r(x) = h(x)/(x-\alpha).$$

Now, the last assertion 'is obtained by',

f is separable 'if and only if' f has no multiple root 'if and only if' $\deg(f(x), D_x f(x)) = 0$ 'if and only if' $(f(x), D_x f(x)) = 1$.

Thm. A field of characteristic 0 is perfect.

Proof. Let $f(x) \in F[x]$ be an irreducible polynomial. If $D_x f(x)$ is not zero, then $\deg D_x f < \deg f$.

Therefore, $(f, D_x f) = 1$, because if $(f, D_x f) \neq 1$, then it is contradiction to f is irreducible. So it is separable.

Lemma. A field of characteristic p is perfect iff every element of F has a p th root in F .

Proof. Let F be a field of characteristic p .

Suppose that F is perfect. Let $\alpha \in F$ be a non-zero, non-1. put $f(x) = x^p - \alpha$.

If $f(x)$ has a root in F , then there is nothing to prove.

Assume f has no root in F . Observe that in splitting field K of $f(x)$, put $\beta \in K$ s.t.

$$f(\beta) = \beta^p - \alpha = 0.$$

And, $f(x) = x^p - \alpha = x^p - \beta^p = (x - \beta)^p$ in K . But, if f is reducible in F , then for some $1 \leq k < p$, $f(x) = (x - \beta)^{p-k} \cdot (x - \beta)^k$ and $(x - \beta)^{p-k}, (x - \beta)^k \in F[x]$.

That is, $(-\beta)^k \in F$. Clearly, $(k, p) = 1$, thus $ku + pv = 1$ for some $u, v \in \mathbb{Z}$.

Now, $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$. It is contradiction. But, if f is irreducible then $D_x f = p \cdot x^{p-1} = 0$ implies contradiction with F is perfect. Now, α has a p th root in F .

Conversely, let $f(x) \in F[x]$ be an irreducible. Assume it is inseparable.

Denote $f(x) = a_n x^n + \dots + a_1 x + a_0$. Since $f(x)$ is irreducible and inseparable,

$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$ must be zero, that is,

$$a_k \neq 0 \Leftrightarrow k \in (p)$$

Therefore, we can write

$$f(x) = b_m x^{mp} + b_{m-1} x^{(m-1)p} + \dots + b_1 x^p + b_0$$

Now, by the assumption, each b_i has p th root c_i , thus

$$f(x) = c_m^p \cdot (x^m)^p + c_{m-1}^p \cdot (x^{m-1})^p + \dots + c_0^p$$

$$= (c_m \cdot x^m + c_{m-1} \cdot x^{m-1} + \dots + c_0)^p. \text{ But it is reducible. Contradiction.}$$

Corollary. Every finite field is perfect. (char is p on finite thus Auto.)

Thm. Every finite separable extension is simple extension.

Proof. If F is finite, then the finite extension E/F is also finite. Then, E^X is cyclic, therefore $E = F(\alpha)$ where α is generator of E^X .
Now, assume that F is infinite. Since E/F is finite,

$$E = F(\alpha_1, \dots, \alpha_r) \text{ where } \alpha_1, \dots, \alpha_r \in E.$$

Now, enough to show $F(\beta, \gamma) = F(\alpha)$ for some $\alpha \in E$, and after is solved by the induction.

Let $\beta_1, \dots, \beta_n$ be distinct zeros of $\text{irr}(\beta, F)$, and

$\gamma_1, \dots, \gamma_m$ be distinct zeros of $\text{irr}(\gamma, F)$, in $\bar{F}$. ($\beta_1 = \beta$, $\gamma_1 = \gamma$).

Since E/F is separable, each zeros are multiplicity of 1.

Take $a \in F$ s.t. $a \neq \frac{\beta_i - \beta}{\gamma - \gamma_j}$ for all $1 \leq i \leq n$, $2 \leq j \leq m$, since F is infinite.

Put $\alpha = \beta + a \cdot \gamma$. Then, for all $(i, j) \in \{1, \dots, n\} \times \{2, \dots, m\}$,

$$\alpha = \beta + a \gamma \neq \beta_i + a \gamma_j, \text{ or } \alpha - a \gamma_j \neq \beta_i.$$

Observe that: $\text{irr}(\beta, F)(\alpha - a\gamma) \in F(\alpha)[x]$, and

$$\text{irr}(\beta, F)(\alpha - a\gamma) = \text{irr}(\beta, F)(\beta) = 0.$$

But, $\text{irr}(\beta, F)(\alpha - a\gamma_j) \neq 0$ for all $2 \leq j \leq m$,

That is, $\text{irr}(\beta, F)(\alpha - a\gamma)$ and $\text{irr}(\beta, F)(x)$ have a common factor $(x - \gamma)$ in $F(\alpha)$, thus $\gamma \in F(\alpha)$, so $\beta = \alpha - a\gamma \in F(\alpha)$. ~~so~~

Γ_{Norm}

Let K be a finite normal extension of a field F . Define a Norm

$$N_{K/F}: K^{\times} \rightarrow F^{\times}; \alpha \mapsto \prod_{\sigma \in G(K/F)} \sigma(\alpha)$$

and trace

$$\text{Tr}_{K/F}: K \rightarrow F; \alpha \mapsto \sum_{\sigma \in G(K/F)} \sigma(\alpha)$$

are well-defined. Because for any $\mu \in G(K/F)$,

$$\mu(N_{K/F}(\alpha)) = \mu\left(\prod \sigma(\alpha)\right) = \prod (\mu \circ \sigma)(\alpha) = \prod \sigma(\alpha) = N_{K/F}(\alpha)$$

Therefore, $N_{K/F}(\alpha) \in K_{G(K/F)} = F$, for any $\alpha \in K$.

Change the prod $\prod$ to sum $\sum$, we obtain the result for Tr .

Moreover, each $N_{K/F}$ and $\text{Tr}_{K/F}$ preserve the operation, respectively:

$$N(\alpha\beta) = N(\alpha)N(\beta) \quad \text{and} \quad \text{Tr}(\alpha+\beta) = \text{Tr}(\alpha) + \text{Tr}(\beta)$$

Def. Let K be a field, and $K_1, K_2 \leq K$ be subfields.

A smallest subfield containing K_1 and K_2 is called *compositum* (or *join*)

$$K_1 \vee K_2 \stackrel{\text{def}}{=} \bigcap_{\substack{E \leq K \\ K_1, K_2 \leq E}} E$$

Lemma. Let E/F be an extension field, and let $\alpha_1, \dots, \alpha_n \in E$.

$$F(\alpha_1) \vee \dots \vee F(\alpha_n) = F(\alpha_1, \dots, \alpha_n).$$

Proof. Recall that: $F(\alpha_1, \dots, \alpha_n) = \bigcap_{\substack{F \leq H \leq E \\ \alpha_1, \dots, \alpha_n \in H}} H$.

And, for any $F \leq H \leq E$, $\alpha_i \in H \Leftrightarrow F(\alpha_i) \leq H$. Thus proved.

Thm. Let $F \leq E \leq \bar{F}$ be a tower of fields.

E is splitting field over F if and only if every automorphism of $\bar{F}$ leaving F maps E onto E .

Proof. Suppose that E is a splitting field over F in $\bar{F}$ of $\{f_i(x) \mid i \in I\}$.

Define $S_i = \{d \in \bar{F} \mid f_i(d) = 0\}$, for each $i \in I$. Take $S = \bigcup_{i \in I} S_i$.

Then, by the Lemma, $E = \bigvee_{d \in S} F(d)$.

Now, for any Automorphism $\sigma: \bar{F} \rightarrow \bar{F}$ leaving F fixed,

$$\sigma[E] = \sigma\left[\bigvee_{d \in S} F(d)\right] = \bigvee_{d \in S} \sigma[F(d)] = \bigvee_{d \in S} F(\sigma(d)) = \bigvee_{d \in S} F(d) = E.$$

σ leaves F fixed σ permutes on S .

$$\begin{aligned} \mathcal{Q}[K_1 \vee K_2] &= \mathcal{Q}\left[\bigcap_{\substack{E \leq K \\ K_1, K_2 \leq E}} E\right] = \bigcap_{\substack{E \leq K \\ K_1, K_2 \leq E}} \mathcal{Q}[E] = \mathcal{Q}[K_1] \vee \mathcal{Q}[K_2] \\ &\hookrightarrow \mathcal{Q}[K_i] \leq \mathcal{Q}[E] \\ &\Leftrightarrow \mathcal{Q}[K_i] \leq \mathcal{Q}[E] \end{aligned}$$

Thm. Let K be a finite extension of degree n of a finite field F , with p^n elements. Then, $G(K/F) \cong \mathbb{Z}_n$, generated by $\alpha \mapsto \alpha^{p^n}$.

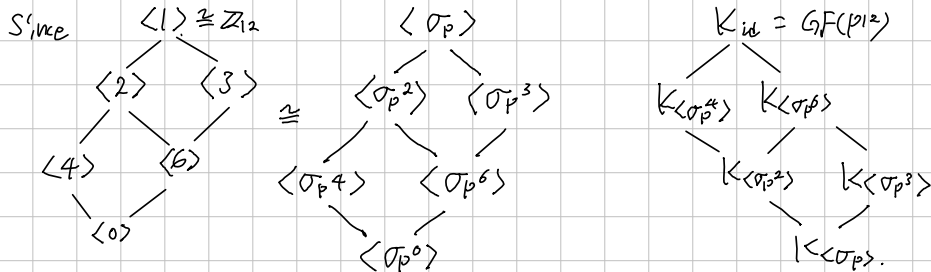
Proof. K is splitting field of $x^{p^n} - x$ over F . And it is finite, separable, thus normal. And, $\sigma(x) = x^{p^n}$ is element in $G(K/F)$, with

$$\sigma^2(x) = x^{p^{2n}}.$$

Since $x^{p^{r_i}} - x$ can have at most r_i zeros, $|\sigma| \geq n$.

And, since $|G(K/F)| = [K:F] = n$, $G(K/F)$ must be cyclic.

Example. $F = \mathbb{Z}_p$, $K = GF(p^2)$. Then, $Gal(K/F) = \langle \sigma \mapsto x^p \rangle \cong \mathbb{Z}_2$.



$$GF(p^r) = \{x \in \overline{\mathbb{Z}_p} \mid x^{p^r} - x = 0\} = K_{\langle \sigma_p^r \rangle}.$$

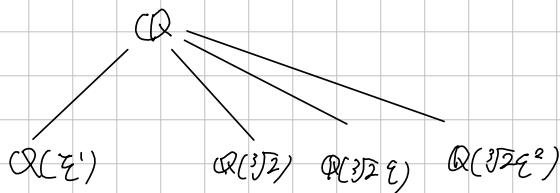
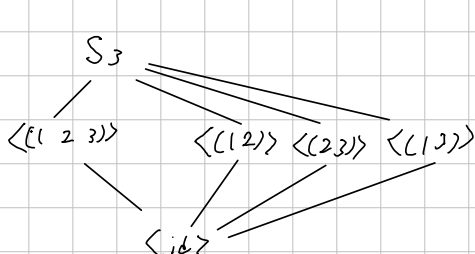
Splitting field of x^3-2 over $\mathbb{Q}$.

Since x^3-2 has the three solutions

$$\sqrt[3]{2}, \sqrt[3]{2} \cdot \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right), \sqrt[3]{2} \cdot \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$$

The splitting field is $K = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$, $[K:\mathbb{Q}] = 6$. Moreover,

The element $\sigma \in \text{Gal}(K/\mathbb{Q})$ conjugates $i\sqrt{3}$ to $\pm i\sqrt{3}$ and $\sqrt[3]{2}$ to $\sqrt[3]{2} \cdot \exp(2\pi i \frac{k}{3})$,
 It is isomorphic to D_6 , or S_3 . Nam, ($k=0,1,2$)



$F = \mathbb{Q}$, K_1 is splitting field of x^3-2 ,
 K_2 // x^3-5 .

Example. Let K be a splitting field of $x^4 - 2$ over $\mathbb{Q}$.

Since the roots of $x^4 - 2$ is $\sqrt[4]{2} \cdot i^k$ ($k = 0, 1, 2, 3$), $K = \mathbb{Q}(\sqrt[4]{2}, i)$.

And, since $[K: \mathbb{Q}] = [\mathbb{Q}(i: \mathbb{Q}(\sqrt[4]{2}))][\mathbb{Q}(\sqrt[4]{2}): \mathbb{Q}] = 2 \cdot 4 = 8$, $|G(K/\mathbb{Q})| = 8$.

Moreover, an element in $G(K/\mathbb{Q})$ conjugates i to $\pm i$, and

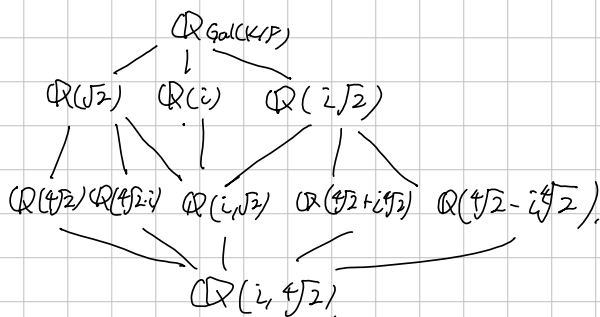
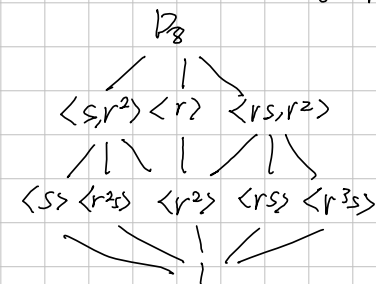
" $\sqrt[4]{2}$ to $\sqrt[4]{2} \cdot i^k$ ($k = 0, 1, 2, 3$).

Let $s \in G(K/\mathbb{Q})$ s.t. $s(i) = -i$, the others fixed.

Let $r \in G(K/\mathbb{Q})$ s.t. $r(\sqrt[4]{2}) = \sqrt[4]{2}i$, the others fixed.

Then, Now $G(K/\mathbb{Q}) = \{s^i \circ r^j \mid i = 0, 1, j = 0, 1, 2, 3\} \cong D_8$.

Moreover, $s^2 = r^4 = (s \circ r)^2 = \text{id}$.



Example. Let K be a splitting field of x^4+1 over $\mathbb{Q}$.

Since x^4+1 is irreducible over $\mathbb{Q} \Leftrightarrow (x+1)^4+1$ is irreducible over $\mathbb{Q}$,
but, by the Eisenstein criterion, $x^4+\binom{4}{1}x^3+\binom{4}{2}x^2+\binom{4}{3}x+1+1$ is irreducible,
with respect to $p=2$. Meanwhile, a solution of $x^4=-1$ is

$$x^4 = \exp(\pi i), \text{ thus } x = \exp\left(\frac{k}{4}\pi i\right), k=1,3,5,7$$

$$\text{and } \exp\left(\frac{1}{4}\pi i\right) = \frac{1+i}{\sqrt{2}}. \text{ That is, } K = \mathbb{Q}(\sqrt{2}i).$$

$$\text{Gal}(K/F) \cong V, \text{ Klein-4 group.}$$

Def. Let F be a field, and let $y_1, \dots, y_n$ be indeterminates

Denote $E = F(y_1, \dots, y_n)$. For each $\sigma \in S_n$, define

$$\bar{\sigma}: E \rightarrow E; \frac{f(y_1, \dots, y_n)}{g(y_1, \dots, y_n)} \mapsto \frac{f(y_{\sigma(1)}, \dots, y_{\sigma(n)})}{g(y_{\sigma(1)}, \dots, y_{\sigma(n)})}$$

It is Automorphism of E . And, define $\bar{S}_n = \{ \bar{\sigma} : E \rightarrow E \mid \sigma \in S_n \}$.

Now, $E_{\bar{S}_n}$ is called the symmetric functions.

For each $i = 1, \dots, n$,

$$s_i = \sum_{1 \leq a_1 \leq \dots \leq a_i \leq n} y_{a_1} \cdot y_{a_2} \cdot \dots \cdot y_{a_i}$$

Thm. Let $E = F(s_1, \dots, s_n)$, $K = F(y_1, \dots, y_n)_{\bar{S}_n}$. Then, $E = K$

Proof. Let $f(x) = \prod_{i=1}^n (x - y_i)$ over E , and since $\deg f = n$,

$$[F(y_1, \dots, y_n) : E] \leq n!$$

And, since $|\bar{S}_n| = n!$, $E \leq K$.

$$n! \leq [F(y_1, \dots, y_n) : K] \leq [F(y_1, \dots, y_n) : E] \leq n!$$

Def. K/F is called an Extension of F by radicals if:

for some $\alpha_1, \dots, \alpha_n \in K$, $K = F(\alpha_1, \dots, \alpha_n)$ s.t. $\alpha_i^{n_i} \in F$ for some $n_i \in \mathbb{N}$.

A finite $F[x]$ is solvable by radicals over F if: splitting field of f is contained in

Thm. Let F be a field of characteristic 0, and let $a \in F$.

If K is the splitting field of $X^n - a$ over F , then $G(K/F)$ is solvable.

Proof. Suppose that F contains all the n th roots of unity.

Let ζ be a primitive n th roots of unity. Then,

$$\{\zeta^k \mid k=0,1,\dots,n-1\}$$

are n th roots of unity. If $\beta \in F$ is a zero of $X^n - a$, then

$$\{\zeta^k \beta \mid k=0,1,\dots,n-1\}$$

are zeros of $X^n - a$. By assumption, $K = F(\beta)$, $\sigma \in G(K/F)$ is determined by

the value of $\sigma(\beta)$. Let $\tau, \sigma \in G(K/F)$ s.t. $\tau(\beta) = \zeta^i \beta$, $\sigma(\beta) = \zeta^j \beta$.

Then, $\tau \circ \sigma = \sigma \circ \tau$, $G(K/F)$ is Abelian, thus solvable.

Suppose that F does not contain a primitive n th root of unity.

Let ζ be a primitive n th root of unity, and β be a zero of $X^n - a$.

Since β and $\zeta\beta$ both are zero of $X^n - a$, $\xi = (\zeta\beta)/\beta \in K$.

Now, similarly again, $G(F(\zeta)/F)$ is abelian, and since $F(\zeta)$ is Normal of F ,

$$G(K/F)/G(K/F(\zeta)) \cong G(F(\zeta)/F)$$

Now, $\{id\} \trianglelefteq G(K/F(\zeta)) \trianglelefteq G(K/F)$ have a composition series of abelian quotients, being $G(K/F(\zeta))$ is abelian. Finally $G(K/F)$ is solvable.

Problems.

Let K be the splitting field of x^2+x+1 over $\mathbb{Z}_2$. Is K an extension of $\mathbb{Z}_2$ by radicals?

Proof. Since $x^3-1=(x-1)(x^2+x+1)$, if $\beta \in \overline{\mathbb{Z}_2}$ is root of x^2+x+1 , $\beta^3=1 \in \mathbb{Z}_2$.

Therefore, $K=\mathbb{Z}_2(\beta)$ is an Extension by radical